

VISTA

Querying, Explaining, and Steering World Models Through Language

Extended Abstract

World models are becoming the **predictive substrate of embodied AI**. They can forecast trajectories, simulate futures, estimate uncertainty, and support planning. Yet they remain largely **silent simulators**: users can observe their outputs, but cannot easily ask **why**, explore **what if**, inspect **uncertainty**, or steer future rollouts.

We introduce **VISTA**, a framework for turning predictive world models into **language-addressable simulators**. VISTA treats language not as a captioning layer, but as an **execution interface over world-model states**. A user instruction is mapped to a structured **world operation**: query a future, explain a prediction, intervene on a scene, steer a rollout, extract a plan, or inspect a latent assumption.

VISTA is built as a modular system: a frozen world model generates **future rollouts, occupancy maps, uncertainty estimates, trajectories, and latent states**; a trainable **world-language adapter** converts these into language-addressable tokens; and an instruction-tuned LLM produces grounded responses or control signals. This avoids expensive model merging while allowing VISTA to operate across V-JEPA-style latent predictors, generative video world models, trajectory forecasters, and embodied planners.

Table 1: VISTA Instruction Taxonomy

Family	Instruction Type	Example Query	World Operation	Example Output
QUERY	Future prediction	<i>What happens next?</i>	Roll out future states	Pedestrian crosses; car slows
QUERY	Long-horizon forecast	<i>What will the scene look like in 5 seconds?</i>	Multi-step simulation	Future video/trajectory rollout
QUERY	Multi-future prediction	<i>Show three plausible futures.</i>	Multimodal rollout	Cross / wait / step back
QUERY	Occupancy forecast	<i>Where will the cyclist be after 2 seconds?</i>	Future occupancy prediction	Heatmap over right lane
QUERY	Trajectory forecast	<i>Predict the pedestrian path.</i>	Motion forecasting	Path overlay
QUERY	Scene evolution	<i>How will traffic evolve?</i>	Multi-agent rollout	Queue forms near intersection
EXPLAIN	Prediction explanation	<i>Why do you predict braking?</i>	Attribution analysis	Lead vehicle slows; headway shrinks
EXPLAIN	Evidence extraction	<i>What evidence supports this prediction?</i>	Evidence localization	Gaze, speed, lane position
EXPLAIN	Uncertainty explanation	<i>Why are you uncertain?</i>	Uncertainty decomposition	Occlusion + competing futures
EXPLAIN	Causal analysis	<i>What caused the lane-change prediction?</i>	Causal factor extraction	Slow truck ahead
EXPLAIN	Belief explanation	<i>Why do you think the cyclist will yield?</i>	Latent assumption summary	Narrow gap + oncoming traffic
OCCLUSION	Hidden-state query	<i>Where is the cyclist behind the bus?</i>	Hidden-state inference	Occupancy distribution
OCCLUSION	Object permanence	<i>Is the pedestrian still present?</i>	Persistence reasoning	Yes, confidence 0.92
OCCLUSION	Reappearance prediction	<i>Where will the cyclist emerge?</i>	Occluded rollout	Right side of bus
INTENT	Crossing intent	<i>Is the pedestrian preparing to cross?</i>	Intent estimation	Crossing probability 0.76
INTENT	Directional intent	<i>Which way will the cyclist turn?</i>	Intent distribution	Left turn 0.63
INTENT	Goal inference	<i>Which vehicle intends to merge?</i>	Goal estimation	Vehicle #3
RISK	Risk attribution	<i>Which actor contributes most uncertainty?</i>	Uncertainty attribution	Cyclist dominates variance
RISK	Collision analysis	<i>What is the most dangerous future?</i>	Worst-case rollout	Near-collision branch
RISK	Safety assessment	<i>Which interaction is most risky?</i>	Conflict analysis	Pedestrian-vehicle crossing
COUNTERFACTUAL	Actor intervention	<i>What if the pedestrian starts running?</i>	Modify actor dynamics	Earlier crossing
COUNTERFACTUAL	Speed intervention	<i>What if the cyclist brakes?</i>	Intervention rollout	Vehicle no longer yields
COUNTERFACTUAL	Context intervention	<i>What if it starts raining?</i>	Environment-conditioned rollout	Slower traffic; wider uncertainty
SOCIAL	Role prior probe	<i>What if this rider is a delivery worker?</i>	Role-conditioned rollout	Higher gap acceptance
SOCIAL	Vulnerability probe	<i>What if this pedestrian is a child?</i>	Identity-conditioned rollout	Larger trajectory spread

Family	Instruction Type	Example Query	World Operation	Example Output
SOCIAL	Age prior probe	<i>What if this pedestrian is elderly?</i>	Actor-prior intervention	Slower crossing prediction
SOCIAL	Cultural prior probe	<i>Would this crossing differ in India?</i>	Context-prior rollout	More negotiation; wider occupancy
STEER	Safety steering	<i>Generate the safest future.</i>	Safety-constrained rollout	Conservative trajectories
STEER	Worst-case steering	<i>Generate the most dangerous future.</i>	Adversarial rollout	Collision-focused branch
STEER	Conservative steering	<i>Assume maximum caution.</i>	Low-risk rollout	Earlier braking/yielding
STEER	Belief steering	<i>Assume the cyclist is distracted.</i>	Prior injection	Increased trajectory variance
PLAN	Action recommendation	<i>What should the ego vehicle do?</i>	Policy extraction	Slow down and yield
PLAN	Risk-minimizing action	<i>Which action minimizes collision risk?</i>	Planning over rollouts	Controlled braking
PLAN	Comparative planning	<i>Compare brake vs continue.</i>	Action comparison	Braking lowers risk by 71%
DEBUG	Failure diagnosis	<i>Why did planning fail?</i>	Failure attribution	Occluded cyclist underestimated
DEBUG	Assumption inspection	<i>Which assumption caused this error?</i>	Belief tracing	Incorrect yielding assumption
DEBUG	Before/after analysis	<i>Show prediction before and after intervention.</i>	Rollout comparison	Side-by-side futures
INSPECT	Belief inspection	<i>What do you believe about this actor?</i>	Prior extraction	High trajectory variability
INSPECT	Predictability ranking	<i>Which actor is least predictable?</i>	Behavioral-prior analysis	Child pedestrian
INSPECT	Vulnerability ranking	<i>Who receives the largest safety margin?</i>	Vulnerability analysis	Wheelchair user
INSPECT	Belief update tracking	<i>What changed after this frame?</i>	Belief differencing	Crossing probability rises

We train VISTA in stages. First, the adapter learns **world-to-language grounding** from rollout-caption pairs. Second, instruction tuning teaches the system to answer VISTA-style queries. Third, lightweight steering heads map language instructions into rollout controls for counterfactual and safety-conditioned simulation. This staged design makes VISTA feasible without full retraining of either the world model or the LLM.

We introduce **VISTA-Bench** to evaluate whether language can faithfully access and manipulate world-model predictions. Metrics include **future-answer correctness**, **rollout-groundedness**, **counterfactual consistency**, **explanation faithfulness**, **uncertainty calibration**, **steering success**, and **planning utility**.

We also release a **Hugging Face demo** where users upload a **10-second video** and ask natural-language questions such as *What happens next?*, *Where is the hidden cyclist?*, *What if the pedestrian runs?*, or *Generate the safest future*. The demo returns **text answers**, **future overlays**, **uncertainty heatmaps**, **trajectory predictions**, and **counterfactual comparisons**.

VISTA reframes world models from passive predictive engines into **interactive reasoning systems**. Future world models should not only predict what happens next; they should explain **why**, explore **alternatives**, expose **uncertainty**, and allow humans to **steer imagined futures through language**.